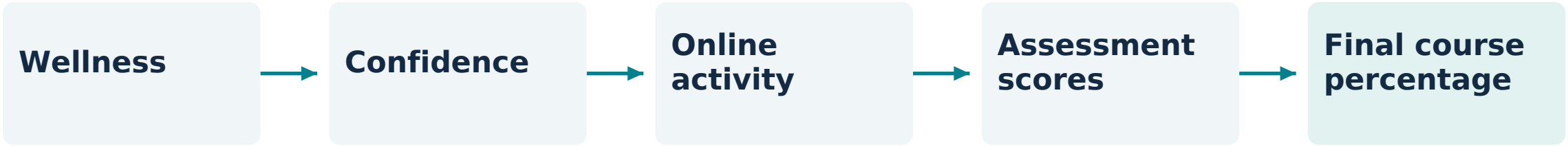
A Bayesian DAG is a map I can learn from

My goal is to estimate a student’s final course percentage and show how uncertain that estimate is.

DAG = boxes and arrows, with no loops

Bayesian = update estimates using data



One path from my larger model; other connections are omitted here.

The arrows are assumptions to examine, not proof of cause and effect.

Next: what information do I put into this map?

I fill the map with four kinds of information

I joined the records into one row for each student in a course term.

191 students • 6 terms • Fall 2024–Spring 2026

Wellness

How the student reported feeling

Motivation

Confidence, course value, barriers and support

Online activity

Page views and recorded participation

Assessment history

Quiz, paper and assignment percentages

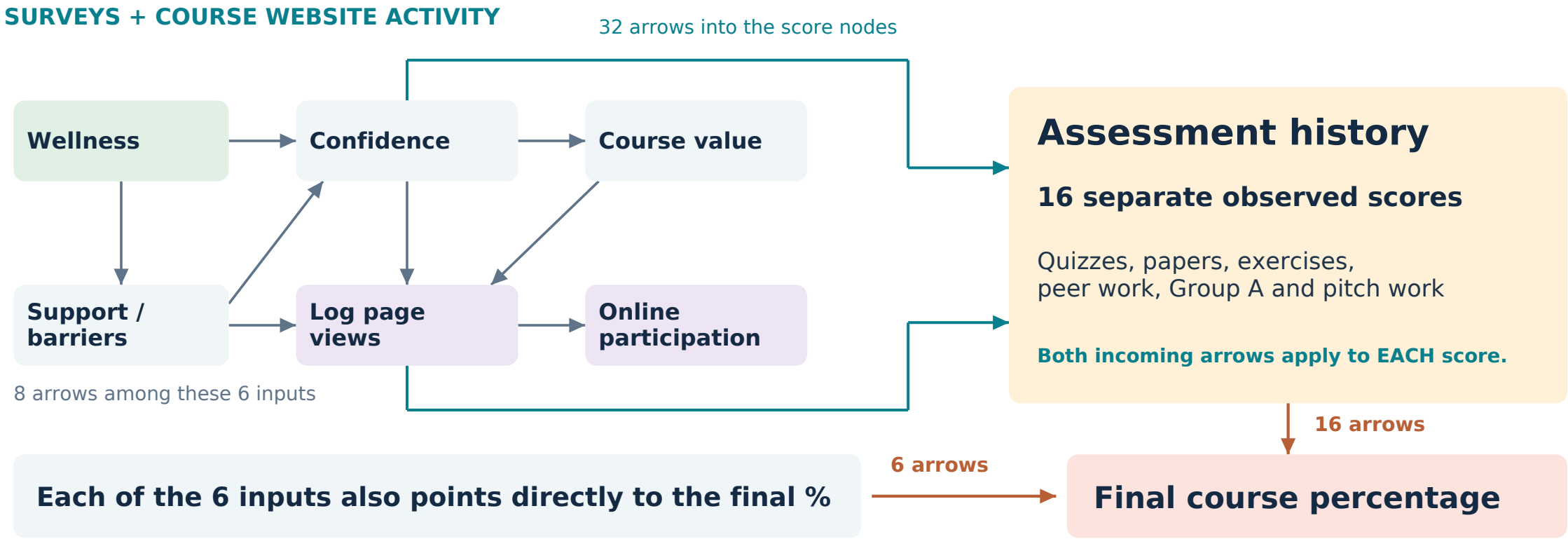
The input sets differ in how many assessment scores they include.

I still need to verify when those scores were available to students.

Next: how are these inputs connected in the original graph?

The original DAG used in my notebook

23 variables and 62 assumed arrows; the assessment variables are grouped here for readability.



Same fitted structure: 8 + 32 + 22 = 62 arrows. Arrow presence is an assumption, not evidence of predictive value.

I learn from past students, then make a forecast

I keep the arrows fixed while the data update the strength of each relationship.

1

Start cautiously

Avoid extreme estimates from limited data.

→

2

Learn from past results

Use earlier students' inputs and known final grades.

→

3

Predict another result

Use the available inputs to estimate a final grade.

One historical example, checked after the final result was known

83.1%

Predicted final result

77.2–89.0%

Reported 90% range

81.0%

Recorded final result

Next: the technical methods used to learn and calculate the forecast.

Technical methods behind my Bayesian DAG

The arrows stay fixed. These methods learn their strengths and turn inputs into a forecast.

METHOD	WHAT I USED	WHY I USED IT
Fixed DAG + linear Gaussian node models	Each value = weighted inputs + Gaussian noise	Explicit assumptions; keep percentage scores continuous
Normal–Inverse-Gamma shrinkage	Bayesian weight / noise estimates Coefficient precisions: 1 vs 10	Limit unstable weights from small samples / overlapping marks
Condition the joint Gaussian on observed inputs	Compute P(final % | inputs) from the fitted joint model	Personalize the forecast using what is known for that student
Marginalize missing evidence	Integrate over unknown inputs at prediction time	Predict when some student inputs are not reported
Final-% mean + 90% interval	Mean μ; Gaussian μ ± 1.645σ Optional past-error rescaling	Show the expected result and an uncertainty range

Training gaps use medians. Main intervals use fitted parameters; they do not include full parameter uncertainty.

I learn from earlier terms and check later students

A forecast is checked against a result that was kept out of model fitting.

Course terms in time order

Fall 2024	Winter 2025	Spring 2025	Fall 2025	Winter 2026	Spring 2026
18 students	27 students	43 students	20 students	40 students	43 students

LEARN: 148 students from the first five terms

CHECK: 43

I chose the model settings using only the earlier terms.

Spring 2026 was kept out of fitting; its results have already been reviewed.

Next: how close were the forecasts, and did the extra inputs help?

The extra inputs have not improved average error yet

All three approaches below are checked on the same 43 Spring 2026 students.

Average error in final percentage points

Lower is better

Same 79.92% forecast for everyone	**8.13**
Use each student’s assessment scores	**2.82**
Also add surveys + online activity	**2.83**

39 of 43

Actual results inside the reported 90% ranges

25.7 points

Largest individual error

Next: compare the same students at the same earlier prediction date.

Next, I will test the extra inputs earlier

Question: do surveys and online activity help before there are many assessment scores?

1

Choose one prediction date

For example, week 4—after verifying what was available.

2

Add one source of information at a time

Grades only → add wellness, motivation, activity, or all three.

3

Look for improvement that holds up

Use the existing later terms; check missing surveys and prediction errors.

My question for you: which prediction date and size of improvement would be useful?